ϵ–transitions Automatons

Computation and Discrete Structures III

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Material adapted from Professor Oscar Bedoya

2026

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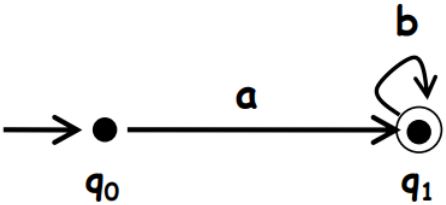
1. Automata with ϵ-Transitions

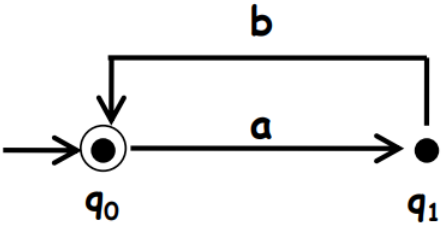
2. Equivalence between NFA-ϵ and NFA (and subsequently DFA)

3. Regular Languages and Finite Automata (Construction)

Automata with ϵ-Transitions

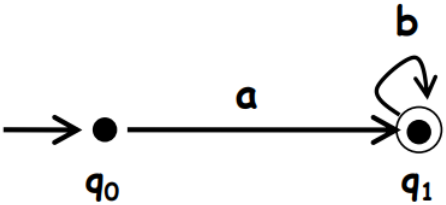
Consider the following two automata that accept ab^* and $(ab)^*$ respectively

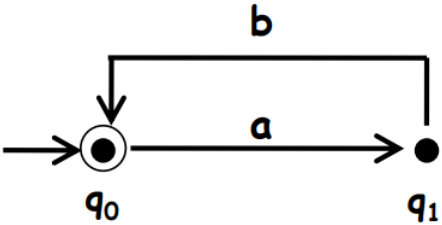




Automata with ϵ-Transitions

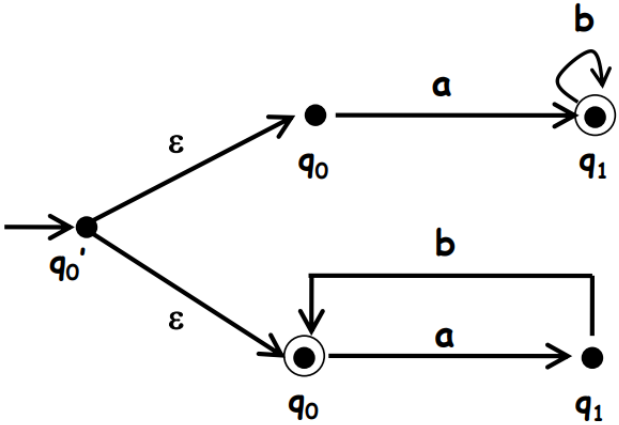
Suppose it is required to design an automaton that accepts $ab^* \cup (ab)^*$





Automata with ϵ-Transitions

Epsilon-transitions can be added from a new initial state q_0' to the initial states of both automata



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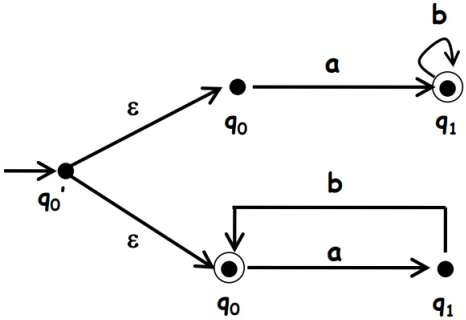
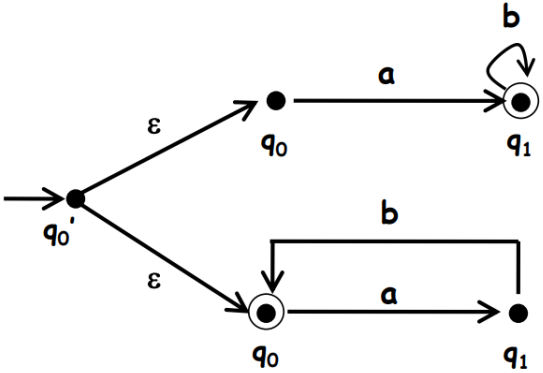
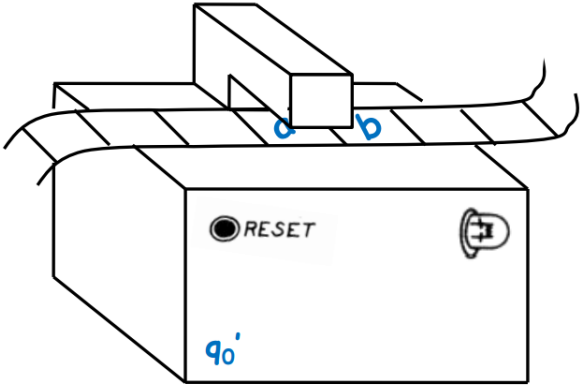


An ϵ-transition indicates that the machine can change its internal state without consuming the input tape symbol

Automata with ϵ-Transitions

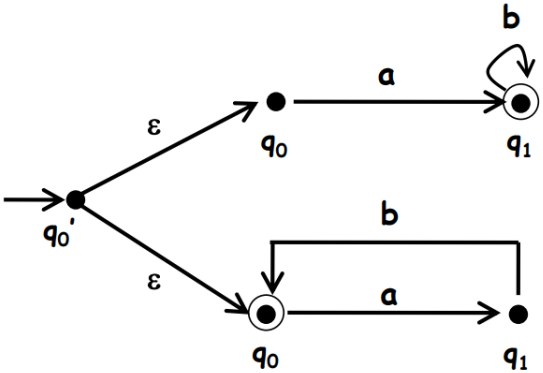
Suppose the string *ab* is read

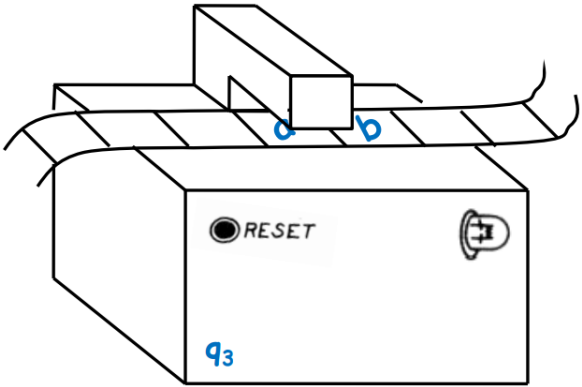




Automata with ϵ-Transitions

The internal state changes without advancing on the tape

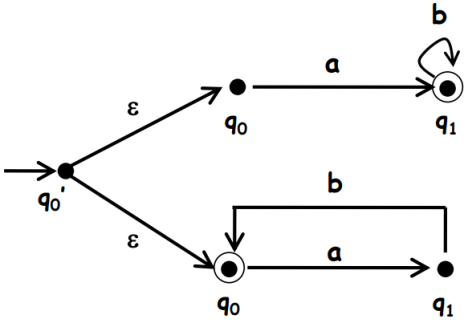




Automata with ϵ-Transitions

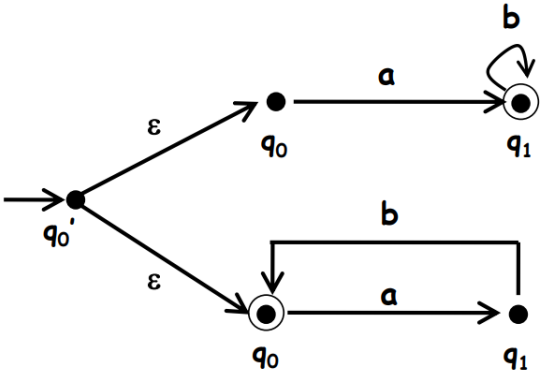
NFA-ϵ

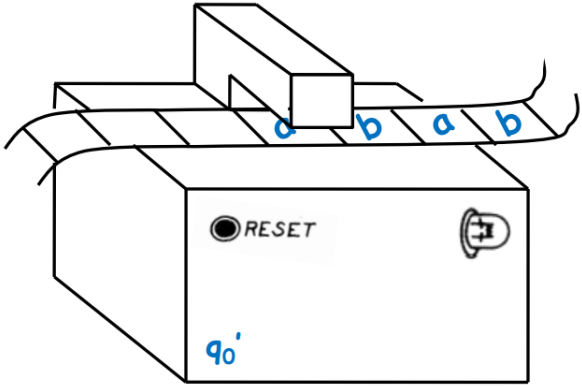
- The resulting automaton is nondeterministic
- Even if the automata were deterministic, ϵ-transitions produce a nondeterministic one



Automata with ϵ-Transitions

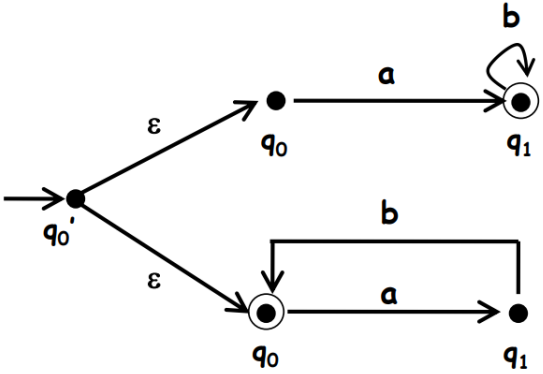
Does the finite automaton accept or reject the string *abab*?

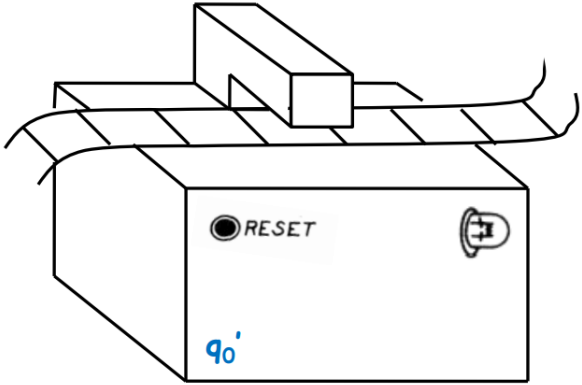




Automata with ϵ-Transitions

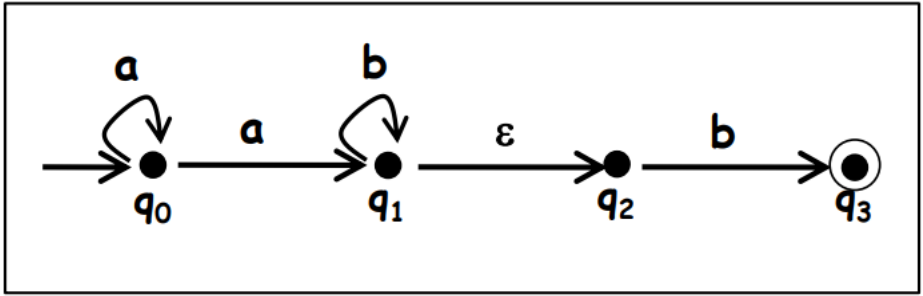
Even the decision to accept the empty string ϵ is nondeterministic





Automata with ϵ-Transitions

Indicate which of the following strings are accepted by the NFA-ϵ: $\epsilon, a, ab, b, ab^{10}$



Automata with ϵ-Transitions

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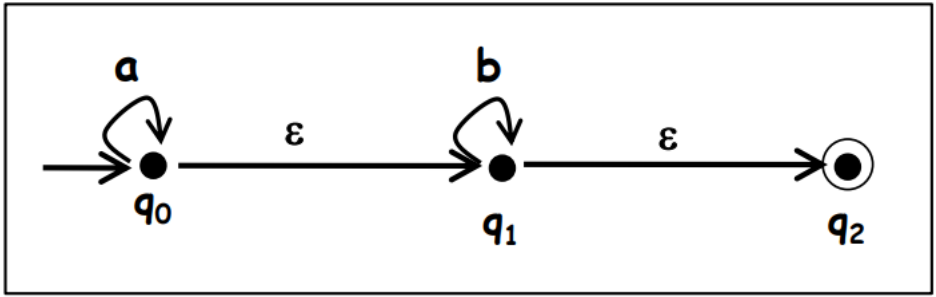
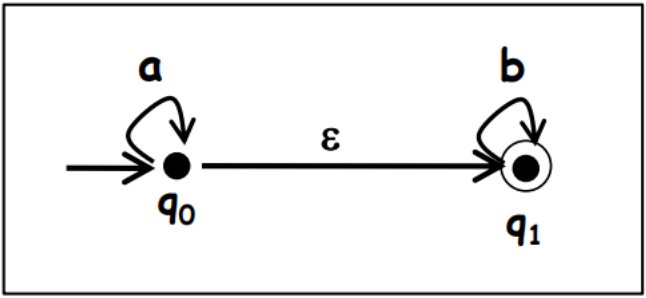


Automata with ϵ–Transitions

Indicate which of the following strings are accepted by the NFA-ϵ: ϵ, a^{10}, $a^{10}b$, ab^{10}, $b^{10}a$



Automata with ϵ–Transitions

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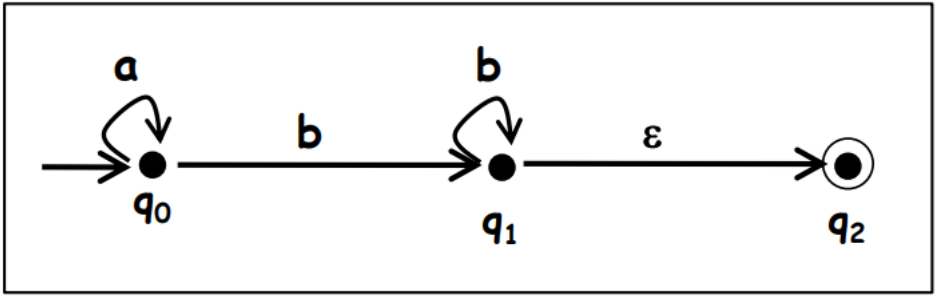

Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

An NFA-ϵ is a collection of five elements:

- An alphabet Σ
- A finite collection of states Q
- An initial state q_0
- A finite collection of accepting states *T*
- A transition relation $\Delta : (Q \times (\Sigma \cup \{\epsilon\})) \Rightarrow 2^Q$

Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

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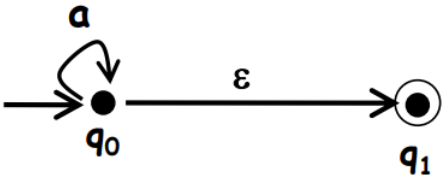
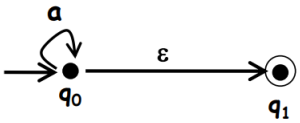


Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

- $\Sigma = \{a\}$
- $Q = \{q_0, q_1\}$
- An initial state q_0
- $T = \{q_1\}$
- $\Delta : (Q \times (\Sigma \cup \{\epsilon\})) \Rightarrow 2^Q$

Δ	a	ε
q_0		
q_1		

ε representa la transición sin consumir símbolo en la cinta



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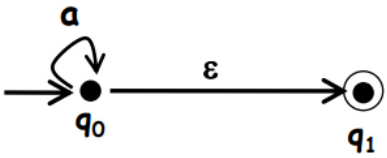
Δ	a	ε
q_0	$\{q_0, q_1\}$	$\{q_0, q_1\}$
q_1	$\varnothing$	$\{q_1\}$

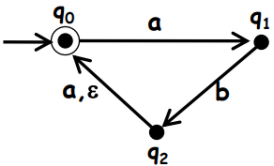


Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

Complete the transition relation:

- $\Sigma = \{a, b\}$
- $Q = \{q_0, q_1, q_2\}$
- An initial state q_0
- $T = \{q_0\}$
- $\Delta : (Q \times (\Sigma \cup \{\epsilon\})) \Rightarrow 2^Q$

Δ	a	b	ε
q_0			
q_1			
q_2			



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- $T = \{q_0\}$
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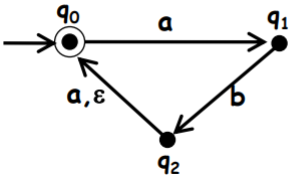
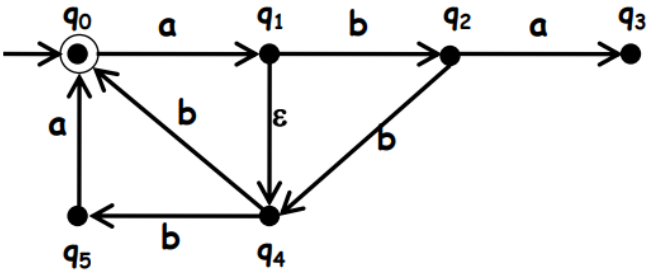
Δ	a	b	ε
q_0	$\{q_1\}$	$\varnothing$	$\{q_0\}$
q_1	$\varnothing$	$\{q_0,q_2\}$	$\{q_1\}$
q_2	$\{q_0,q_1\}$	$\varnothing$	$\{q_0,q_2\}$



Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

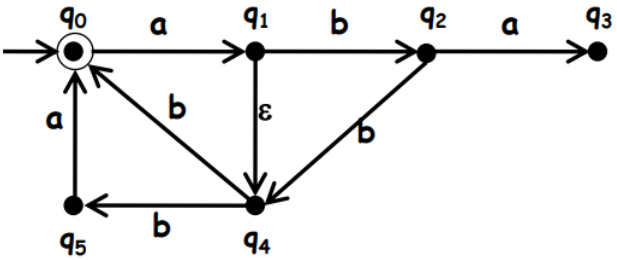
Complete the transition relation for the following NFA-ϵ:

Δ	a	b	ε
q_0			
q_1			
q_2			
q_3			
q_4			
q_5			



Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

Δ	a	b	ε
q_0	$\{q_1, q_4\}$	$\varnothing$	$\{q_0\}$
q_1	$\varnothing$	$\{q_0, q_2, q_5\}$	$\{q_1, q_4\}$
q_2	$\{q_3\}$	$\{q_4\}$	$\{q_2\}$
q_3	$\varnothing$	$\varnothing$	$\{q_3\}$
q_4	$\varnothing$	$\{q_0, q_5\}$	$\{q_4\}$
q_5	$\{q_0\}$	$\varnothing$	$\{q_5\}$



Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

Complete the transition relation for the following NFA-ϵ:

Δ	a	b	ε
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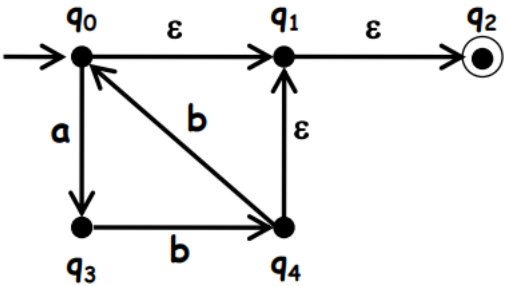
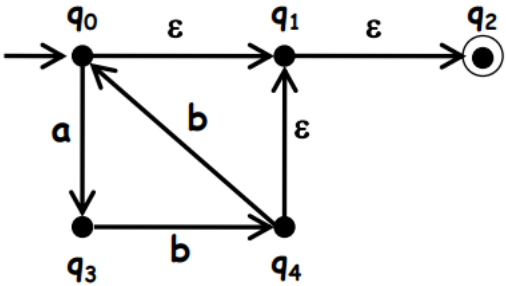
q_1			
q_2			
q_3			
q_4			



Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

Δ	a	b	ε
q_0	$\{q_3\}$	$\varnothing$	$\{q_0,q_1,q_2\}$
q_1	$\varnothing$	$\varnothing$	$\{q_1,q_2\}$
q_2	$\varnothing$	$\varnothing$	$\{q_2\}$
q_3	$\varnothing$	$\{q_1,q_2,q_4\}$	$\{q_3\}$
q_4	$\varnothing$	$\{q_0,q_1,q_2\}$	$\{q_1,q_2,q_4\}$



Contenido

1. Automata with ϵ-Transitions

2. Equivalence between NFA-ϵ and NFA (and subsequently DFA)

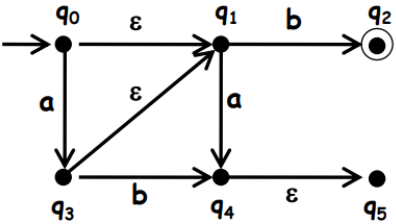
3. Regular Languages and Finite Automata (Construction)

Nondeterministic Finite Automata with ϵ-Tran (NFA-ϵ)

Theorem: NFA-ϵ to NFA

Given an NFA-ϵ M, an equivalent NFA M' (without epsilon transitions) can be constructed

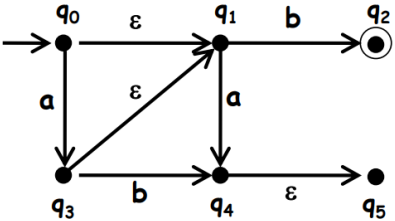
Automata with ϵ-Transitions



Δ	a	b	ε
q_0			
q_1			
q_2			
q_3			
q_4			
q_5			

Δ	a	b
q_0		
q_1		
q_2		
q_3		
q_4		
q_5		

Automata with ϵ-Transitions



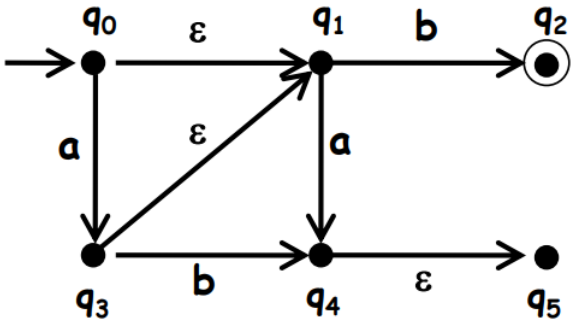
Δ	a	b	ε
q_0			
q_1			
q_2			
q_3			
q_4			
q_5			

Δ	a	b
q_0	$\{q_1,q_3,q_4,q_5\}$	$\{q_2\}$
q_1	$\{q_4,q_5\}$	$\{q_2\}$
q_2	$\varnothing$	$\varnothing$
q_3	$\{q_4,q_5\}$	$\{q_2,q_4,q_5\}$
q_4	$\varnothing$	$\varnothing$
q_5	$\varnothing$	$\varnothing$

Automata with ϵ-Transitions

To find $\Delta(q_0, a)$, starting from q_0, consider:

- States reachable by directly consuming a. $\{q_3\}$
- States reachable first through ϵ-transitions and then by consuming a. $\{q_4\}$
- States reachable after consuming a (previous cases), followed by one or more ϵ-transitions. $\{q_1, q_5\}$



Automata with ϵ-Transitions

To convert an NFA-ϵ to an NFA, the following concepts are used:

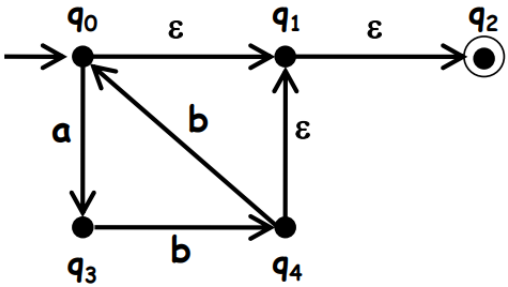
- ϵ-closure of q, $\epsilon - c(q)$
- Direct states, $d(q, \sigma)$

Automata with ϵ-Transitions

ϵ-closure of q

For every state $q \in Q$, the ϵ-closure of q, denoted $\epsilon - c(q)$, is defined as follows:

- $\epsilon - c(q) = \{p \mid p \text{ is a state reachable from } q \text{ without consuming any input symbol}\}$



Automata with ϵ-Transitions

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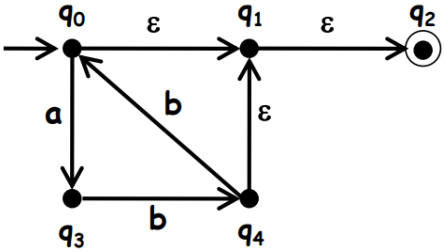

Automata with ϵ–Transitions

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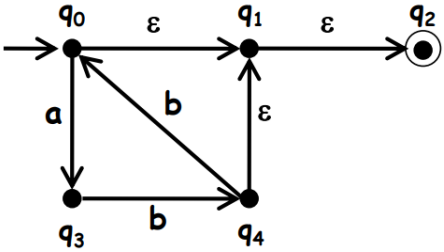
- $\epsilon - c(q_0) = \{q_0, q_1, q_2\}$
- $\epsilon - c(q_1) =$
- $\epsilon - c(q_2) =$
- $\epsilon - c(q_3) =$
- $\epsilon - c(q_4) =$



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- $\epsilon - c(q_1) = \{q_1, q_2\}$
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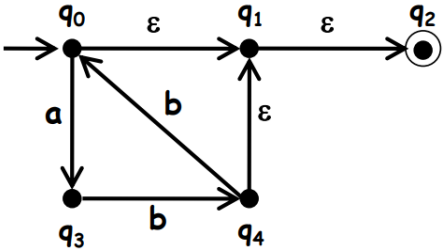
- $\epsilon - c(q_3) = \{q_3\}$
- $\epsilon - c(q_4) = \{q_1, q_2, q_4\}$



Automata with ϵ-Transitions

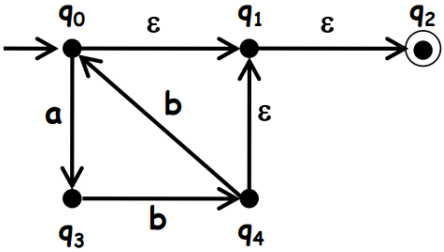
ϵ-closure of q

For every state $q \in Q$, the ϵ-closure of q, denoted $\epsilon - c(q)$, is defined as follows:

- $\epsilon - c(q) = \{p \mid p \text{ is a state reachable from } q \text{ without consuming any input symbol}\}$

$\epsilon - c(q)$

$\epsilon - c(q)$ gives the set of states reachable from q through ϵ-transitions (without consuming any input symbol)



Automata with ϵ–Transitions

Direct States $d(q, \sigma)$

For $q \in Q$ and $\sigma \in \Sigma$, define:

- $d(q, \sigma) = \{p \mid \text{there is a direct transition from } q \text{ to } p \text{ labeled with } \sigma\}$

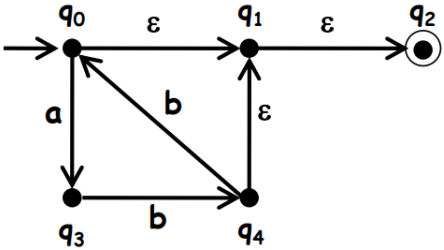
Automata with ϵ–Transitions

Direct States $d(q, \sigma)$

For $q \in Q$ and $\sigma \in \Sigma$, define:

- $d(q, \sigma) = \{p \mid \text{there is a direct transition from } q \text{ to } p \text{ labeled with } \sigma\}$

- $d(q_0, a) =$



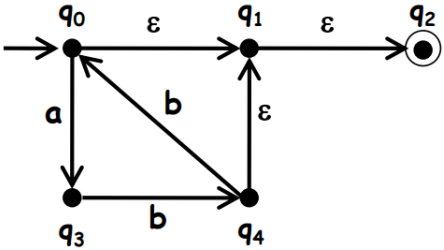
Automata with ϵ–Transitions

Direct States $d(q, \sigma)$

For $q \in Q$ and $\sigma \in \Sigma$, define:

- $d(q, \sigma) = \{p \mid \text{there is a direct transition from } q \text{ to } p \text{ labeled with } \sigma\}$

- $d(q_0, a) = \{q_3\}$
- $d(q_0, b) = \varnothing$
- $d(q_3, a) =$
- $d(q_3, b) =$



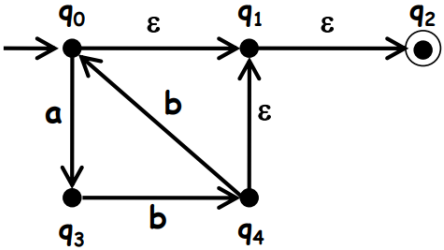
Automata with ϵ–Transitions

Direct States $d(q, \sigma)$

For $q \in Q$ and $\sigma \in \Sigma$, define:

- $d(q, \sigma) = \{p \mid \text{there is a direct transition from } q \text{ to } p \text{ labeled with } \sigma\}$

- $d(q_0, a) = \{q_3\}$
- $d(q_0, b) = \varnothing$
- $d(q_3, a) = \varnothing$
- $d(q_3, b) = \{q_4\}$



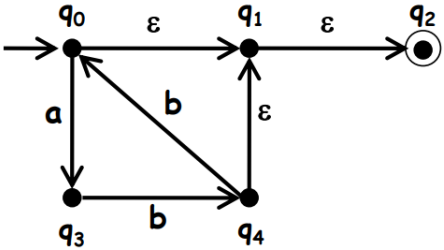
Automata with ϵ–Transitions

Direct States $d(q, \sigma)$

The definition can be extended to apply over sets:

- $d(\{q_1, q_2, ..., q_i\}, \sigma) = d(q_1, \sigma) \cup d(q_2, \sigma) \cup ...d(q_i, \sigma)$

- $d(\{q_3, q_4\}, a) =$
- $d(\{q_3, q_4\}, b) =$
- $d(\{q_0, q_4\}, a) =$
- $d(\{q_0, q_4\}, b) =$



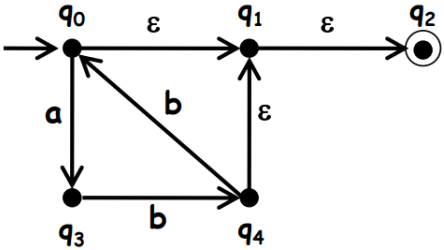
Automata with ϵ-Transitions

Direct States $d(q, \sigma)$

Can be applied over sets:

- $d(\{q_1, q_2, ..., q_i\}, \sigma) = d(q_1, \sigma) \cup d(q_2, \sigma) \cup ...d(q_i, \sigma)$

- $d(\{q_3, q_4\}, a) = \varnothing$
- $d(\{q_3, q_4\}, b) = \{q_4\} \cup \{q_0\} = \{q_0, q_4\}$
- $d(\{q_0, q_4\}, a) =$
- $d(\{q_0, q_4\}, b) =$



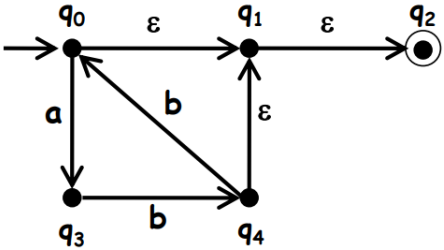
Automata with ϵ–Transitions

Direct States $d(q, \sigma)$

Can be applied over sets:

- $d(\{q_1, q_2, ..., q_i\}, \sigma) = d(q_1, \sigma) \cup d(q_2, \sigma) \cup ...d(q_i, \sigma)$

- $d(\{q_3, q_4\}, a) = \varnothing$
- $d(\{q_3, q_4\}, b) = \{q_4\} \cup \{q_0\} = \{q_0, q_4\}$
- $d(\{q_0, q_4\}, a) = \{q_3\}$
- $d(\{q_0, q_4\}, b) = \{q_0\}$



Automata with ϵ–Transitions

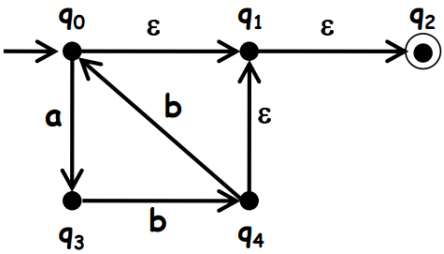
Direct States $d(q, \sigma)$

Can be applied over sets:

- $d(\{q_1, q_2, ..., q_i\}, \sigma) = d(q_1, \sigma) \cup d(q_2, \sigma) \cup ...d(q_i, \sigma)$

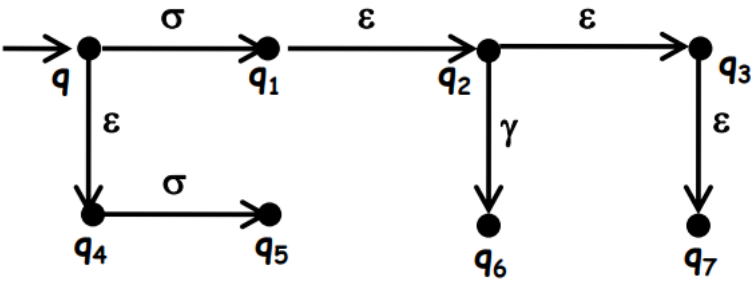
$d(q, \sigma)$

$d(q, \sigma)$ gives the set of states reachable from q through a direct σ-transition



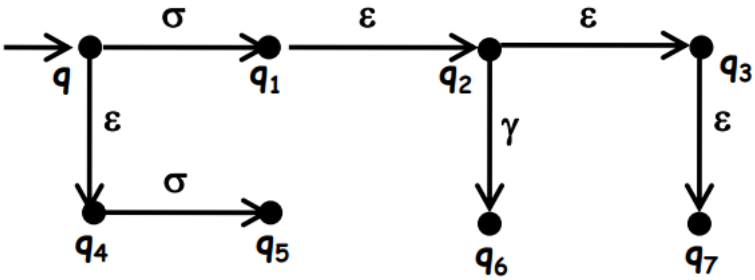
Automata with ϵ-Transitions

Interpret the result of the operation $\epsilon - c(d(q, \sigma))$



Automata with ϵ-Transitions

$\epsilon - c(d(q, \sigma))$ is the set of states reachable from q, first through a σ-transition and then through one or more ϵ-transitions *

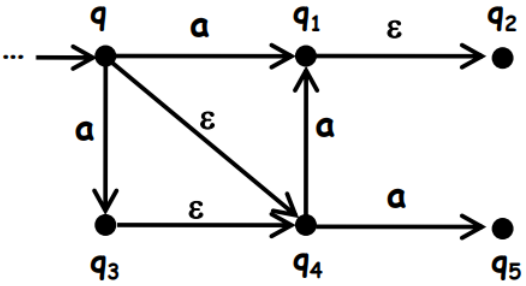


* Every state is reachable from itself without consuming anything; thus, every state is considered reachable from itself via an implicit epsilon self-loop in this type of automaton.

Automata with ϵ-Transitions

$\epsilon - c(d(q, \sigma))$ is the set of states reachable from q, first through a σ-transition and then through one or more ϵ-transitions

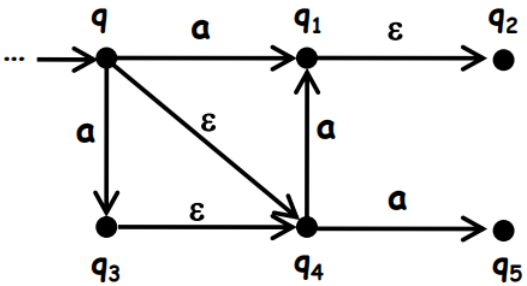
- $\epsilon - c(d(q, a)) =$



Automata with ϵ-Transitions

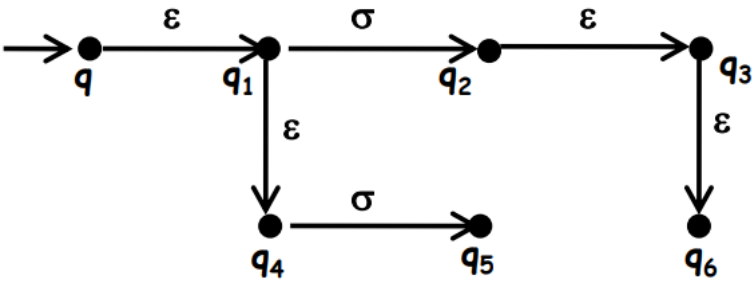
$\epsilon - c(d(q, \sigma))$ is the set of states reachable from q, first through a σ-transition and then through one or more ϵ-transitions

- $d(q, a) = \{q_1, q_3\}$
- $\epsilon - c(d(q, a)) = \{q_1, q_2, q_3, q_4\}$



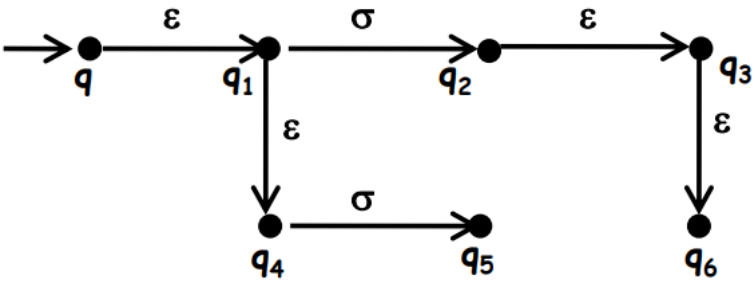
Automata with ϵ-Transitions

Interpret the result of the operation $d(\epsilon - c(q), \sigma)$



Automata with ϵ-Transitions

$d(\epsilon - c(q), \sigma)$ is the set of states reachable from q, first through one or more ϵ-transitions and then a σ-transition

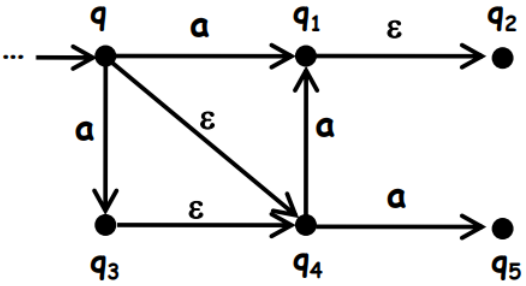


$$d(\epsilon - c(q), \sigma) = d(\{q, q_1, q_4\}, \sigma) = \{q_2, q_5\}$$

Automata with ϵ-Transitions

$\epsilon - c(d(q, \sigma))$ is the set of states reachable from q, first through a σ-transition and then through one or more ϵ-transitions

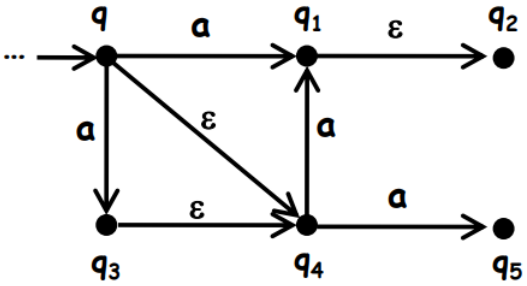
- $d(\epsilon - c(q), a) =$



Automata with ϵ-Transitions

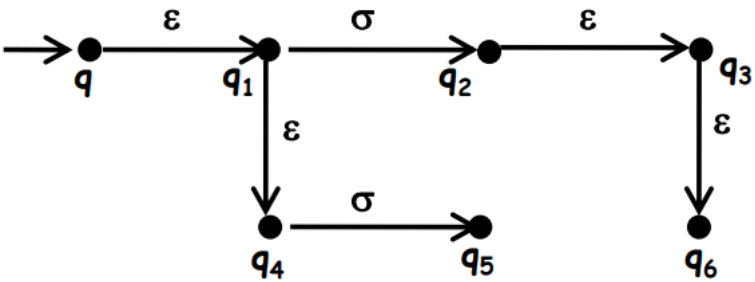
$\epsilon - c(d(q, \sigma))$ is the set of states reachable from q, first through a σ-transition and then through one or more ϵ-transitions

- $\epsilon - c(q) = \{q, q_4\}$
- $d(\epsilon - c(q), a) = \{q_1, q_3, q_5\}$



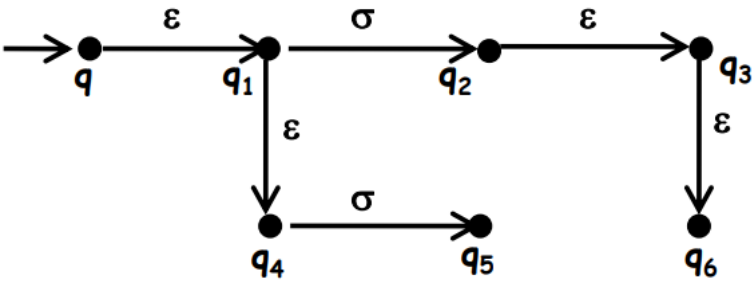
Automata with ϵ-Transitions

Interpret the result of the operation $\epsilon - c(d(\epsilon - c(q), \sigma))$



Automata with ϵ-Transitions

$\epsilon - c(d(\epsilon - c(q), \sigma))$ is the set of states reachable from q, first through one or more ϵ-transitions, then a σ-transition, and then one or more ϵ-transitions

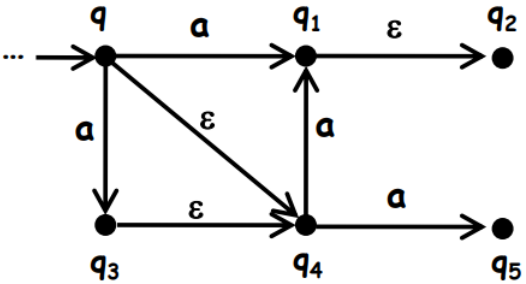


$$\epsilon - c(d(\epsilon - c(q), \sigma)) = \epsilon - c(\{q_2, q_5\}) = \{q_2, q_3, q_5, q_6\}$$

Automata with ϵ-Transitions

$\epsilon - c(d(\epsilon - c(q), \sigma))$ is the set of states reachable from q, first through one or more ϵ-transitions, then a σ-transition, and then one or more ϵ-transitions

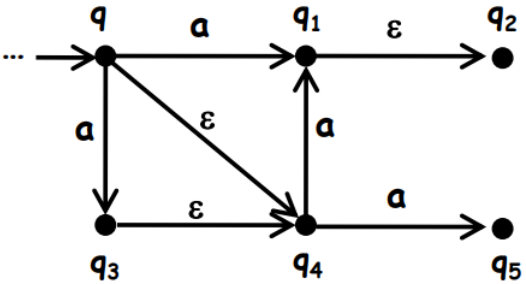
- $\epsilon - c(d(\epsilon - c(q), \sigma)) =$



Automata with ϵ-Transitions

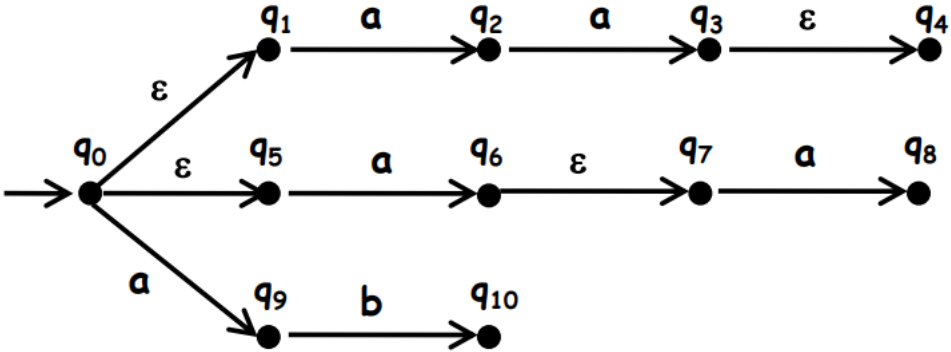
$\epsilon - c(d(\epsilon - c(q), \sigma))$ is the set of states reachable from q, first through one or more ϵ-transitions, then a σ-transition, and then one or more ϵ-transitions

- $\epsilon - c(q) = \{q, q_4\}$
- $d(\epsilon - c(q), a) = \{q_1, q_3, q_5\}$
- $\epsilon - c(d(\epsilon - c(q), \sigma)) = \{q_1, q_2, q_3, q_4, q_5\}$



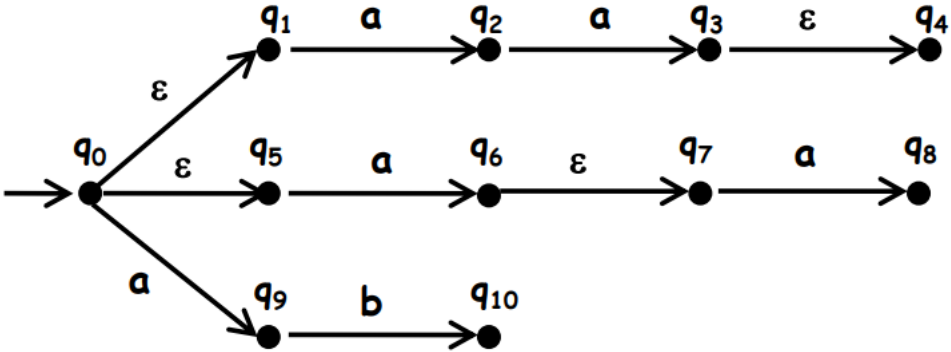
Automata with ϵ-Transitions

- $\epsilon - c(q_0)$
- $d(\epsilon - c(q_0), a)$
- $\epsilon - c(d(\epsilon - c(q_0), a))$



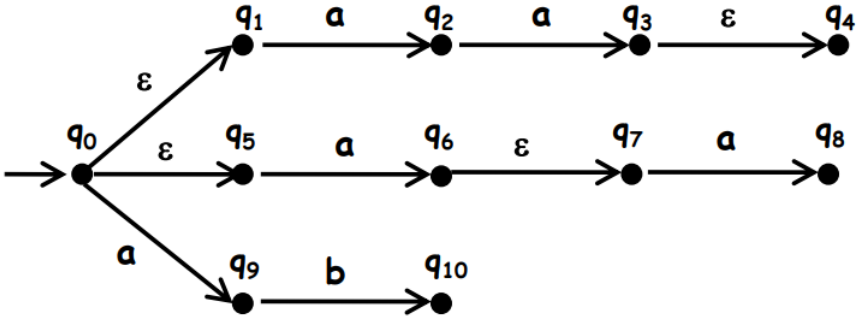
Automata with ϵ-Transitions

- $\epsilon - c(q_0) = \{q_0, q_1, q_5\}$
- $d(\epsilon - c(q_0), a) = \{q_2, q_6, q_9\}$
- $\epsilon - c(d(\epsilon - c(q_0), a)) = \{q_2, q_6, q_7, q_9\}$



Automata with ϵ–Transitions

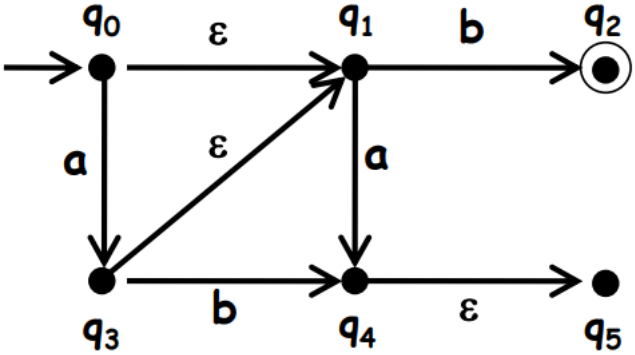
- $\epsilon - c(q_0) = \{q_0, q_1, q_5\}$
- $d(\epsilon - c(q_0), a) = \{q_2, q_6, q_9\}$
- $\epsilon - c(d(\epsilon - c(q_0), a)) = \{q_2, q_6, q_7, q_9\}$



$$\Delta(q_0, a) = \epsilon - c(d(\epsilon - c(q_0), a)) = \{q_2, q_6, q_7, q_9\}$$

Automata with ϵ-Transitions

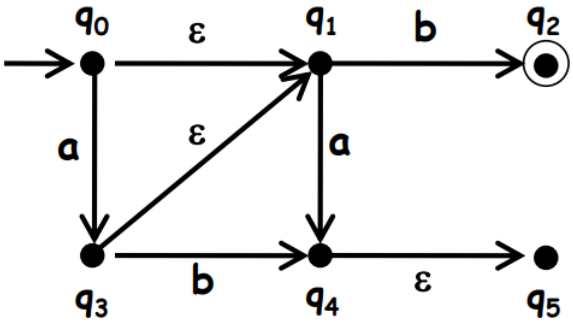
- $\epsilon - c(q_0) =$
- $d(\epsilon - c(q_0), a) =$
- $\epsilon - c(d(\epsilon - c(q_0), a)) =$



Automata with ϵ-Transitions

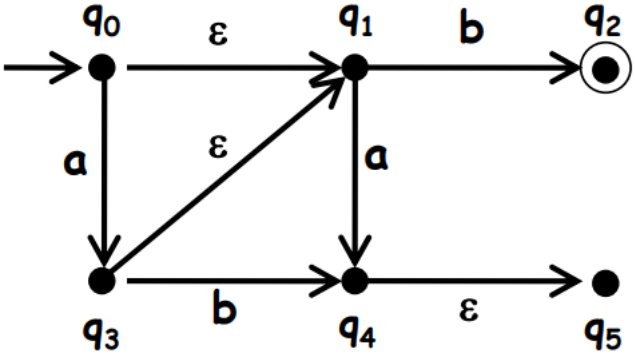
- $\epsilon - c(q_0) = \{q_0, q_1\}$
- $d(\epsilon - c(q_0), a) = \{q_3, q_4\}$
- $\epsilon - c(d(\epsilon - c(q_0), a)) = \{q_1, q_3, q_4, q_5\}$

Therefore, $\Delta(q_0, a) = \{q_1, q_3, q_4, q_5\}$



Automata with ϵ-Transitions

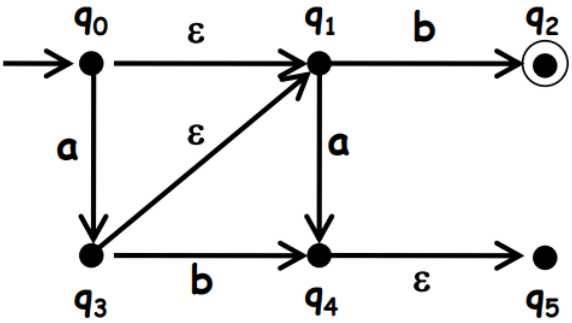
- $\epsilon - c(q_0) =$
- $d(\epsilon - c(q_0), b) =$
- $\epsilon - c(d(\epsilon - c(q_0), b)) =$



Automata with ϵ-Transitions

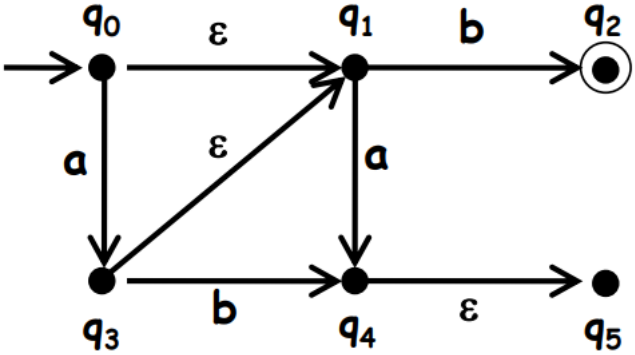
- $\epsilon - c(q_0) = \{q_0, q_1\}$
- $d(\epsilon - c(q_0), b) = \{q_2\}$
- $\epsilon - c(d(\epsilon - c(q_0), b)) = \{q_2\}$

Therefore, $\Delta(q_0, b) = \{q_2\}$



Automata with ϵ-Transitions

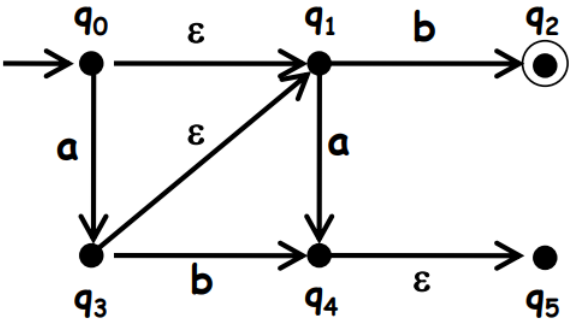
- $\epsilon - c(q_1) =$
- $d(\epsilon - c(q_1), a) =$
- $\epsilon - c(d(\epsilon - c(q_1), a)) =$



Automata with ϵ-Transitions

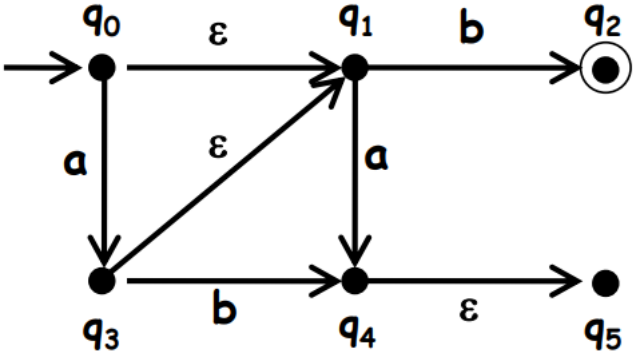
- $\epsilon - c(q_1) = \{q_1\}$
- $d(\epsilon - c(q_1), a) = \{q_4\}$
- $\epsilon - c(d(\epsilon - c(q_1), a)) = \{q_4, q_5\}$

Therefore, $\Delta(q_1, a) = \{q_4, q_5\}$



Automata with ϵ-Transitions

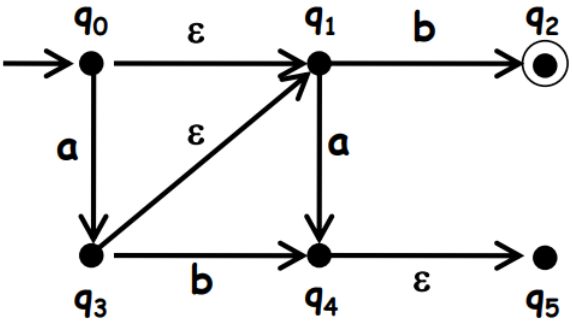
- $\epsilon - c(q_1) =$
- $d(\epsilon - c(q_1), b) =$
- $\epsilon - c(d(\epsilon - c(q_1), b)) =$



Automata with ϵ-Transitions

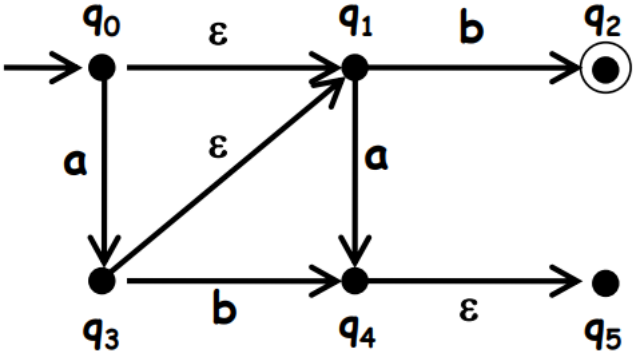
- $\epsilon - c(q_1) = \{q_1\}$
- $d(\epsilon - c(q_1), b) = \{q_2\}$
- $\epsilon - c(d(\epsilon - c(q_1), b)) = \{q_2\}$

Therefore, $\Delta(q_1, b) = \{q_2\}$



Automata with ϵ-Transitions

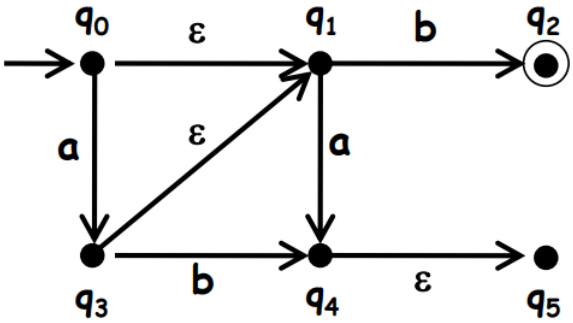
- $\epsilon - c(q_2) =$
- $d(\epsilon - c(q_2), a) =$
- $\epsilon - c(d(\epsilon - c(q_2), a)) =$



Automata with ϵ-Transitions

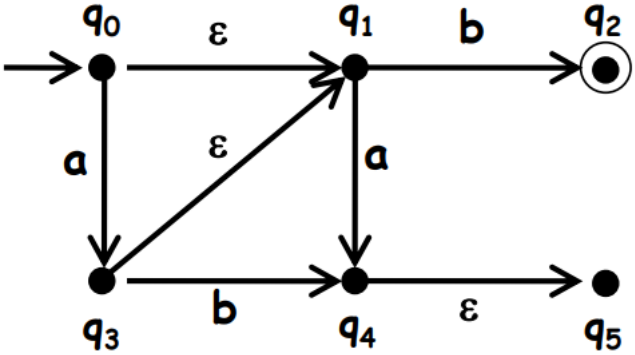
- $\epsilon - c(q_2) = \{q_2\}$
- $d(\epsilon - c(q_2), a) = \varnothing$
- $\epsilon - c(d(\epsilon - c(q_2), a)) = \varnothing$

Therefore, $\Delta(q_2, a) = \varnothing$ and also $\Delta(q_2, b) = \varnothing$



Automata with ϵ-Transitions

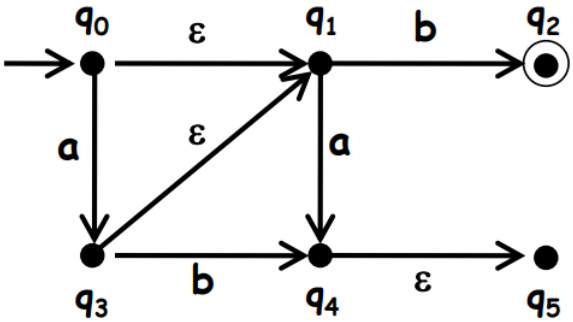
- $\epsilon - c(q_3) =$
- $d(\epsilon - c(q_3), a) =$
- $\epsilon - c(d(\epsilon - c(q_3), a)) =$



Automata with ϵ-Transitions

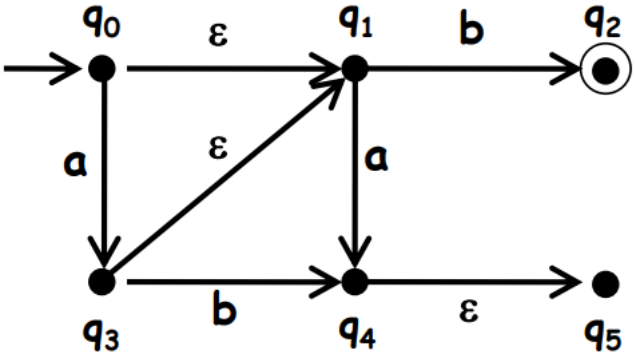
- $\epsilon - c(q_3) = \{q_1, q_3\}$
- $d(\epsilon - c(q_3), a) = \{q_4\}$
- $\epsilon - c(d(\epsilon - c(q_3), a)) = \{q_4, q_5\}$

Therefore, $\Delta(q_3, a) = \{q_4, q_5\}$



Automata with ϵ-Transitions

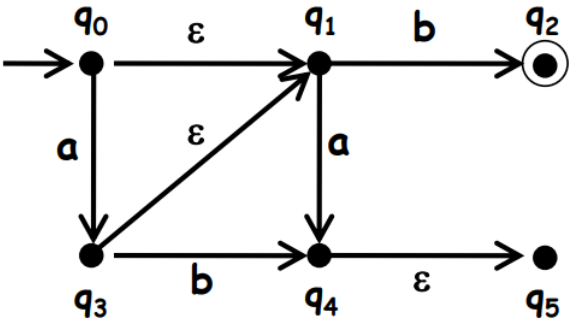
- $\epsilon - c(q_3) =$
- $d(\epsilon - c(q_3), b) =$
- $\epsilon - c(d(\epsilon - c(q_3), b)) =$



Automata with ϵ-Transitions

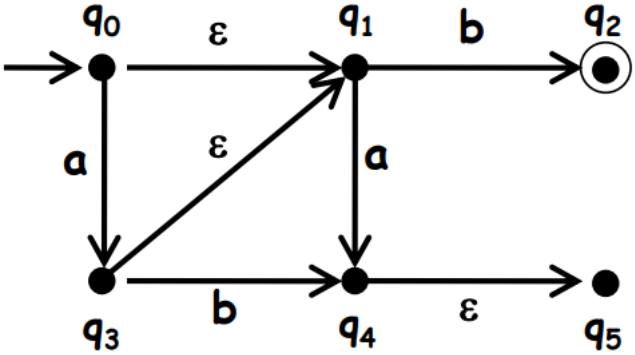
- $\epsilon - c(q_3) = \{q_1, q_3\}$
- $d(\epsilon - c(q_3), b) = \{q_2, q_4\}$
- $\epsilon - c(d(\epsilon - c(q_3), b)) = \{q_2, q_4, q_5\}$

Therefore, $\Delta(q_3, b) = \{q_2, q_4, q_5\}$



Automata with ϵ-Transitions

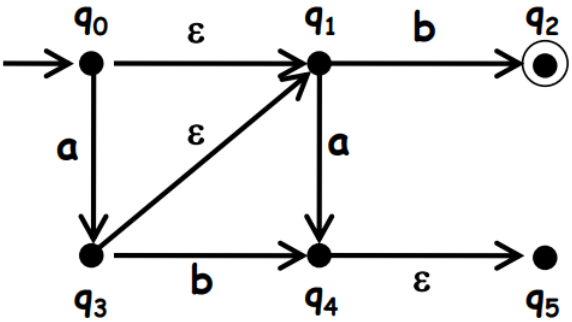
- $\epsilon - c(q_4) =$
- $d(\epsilon - c(q_4), a) =$
- $\epsilon - c(d(\epsilon - c(q_4), a)) =$



Automata with ϵ-Transitions

- $\epsilon - c(q_4) = \{q_4, q_5\}$
- $d(\epsilon - c(q_4), a) = \varnothing$
- $\epsilon - c(d(\epsilon - c(q_4), a)) = \varnothing$

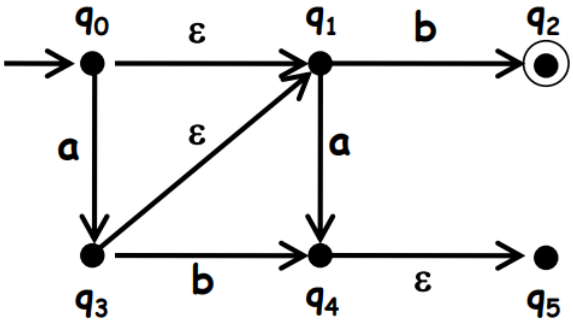
Therefore, $\Delta(q_4, a) = \varnothing$ and also $\Delta(q_4, b) = \varnothing$



Automata with ϵ-Transitions

- $\epsilon - c(q_5) = \{q_5\}$
- $d(\epsilon - c(q_5), a) = \varnothing$
- $\epsilon - c(d(\epsilon - c(q_5), a)) = \varnothing$

Therefore, $\Delta(q_5, a) = \varnothing$ and also $\Delta(q_5, b) = \varnothing$



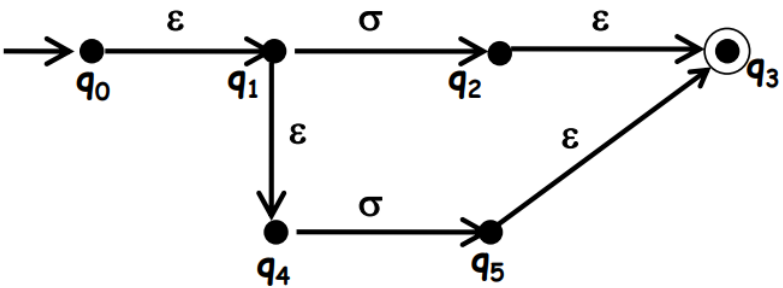
Automata with ϵ-Transitions

The transition relation is defined as follows:

Δ	a	b
q_0	$\{q_1, q_3, q_4, q_5\}$	$\{q_2\}$
q_1	$\{q_4, q_5\}$	$\{q_2\}$
q_2	$\varnothing$	$\varnothing$
q_3	$\{q_4, q_5\}$	$\{q_2, q_4, q_5\}$
q_4	$\varnothing$	$\varnothing$
q_5	$\varnothing$	$\varnothing$

Automata with ϵ–Transitions

To determine the accepting states, ϵ–transitions are taken into account

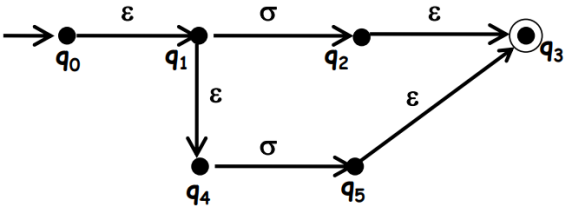


q_3 is an accepting state, but if the computation of a string ends in states q_2 or q_5, it should also be accepted

Automata with ϵ–Transitions

To determine the accepting states, ϵ-transitions are taken into account

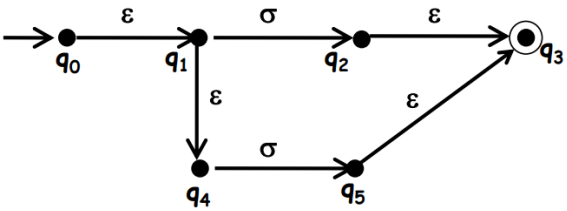
- $\epsilon - c(q_0) = \{q_0, q_1, q_4\}$
- $\epsilon - c(q_1) = \{q_1, q_4\}$
- $\epsilon - c(q_2) = \{q_2, q_3\}$
- $\epsilon - c(q_3) = \{q_3\}$
- $\epsilon - c(q_4) = \{q_4\}$
- $\epsilon - c(q_5) = \{q_3, q_5\}$



Automata with ϵ-Transitions

To determine the accepting states, ϵ-transitions are taken into account

- $\epsilon - c(q_0) = \{q_0, q_1, q_4\}$
- $\epsilon - c(q_1) = \{q_1, q_4\}$
- $\epsilon - c(q_2) = \{q_2, q_3\}$
- $\epsilon - c(q_3) = \{q_3\}$
- $\epsilon - c(q_4) = \{q_4\}$
- $\epsilon - c(q_5) = \{q_3, q_5\}$



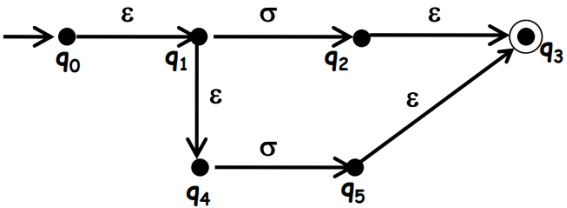
Accepting States T'

The accepting states are those whose ϵ-closure contains q_3; in this case $T' = \{q_2, q_3, q_5\}$

Automata with ϵ–Transitions

To determine the accepting states, ϵ-transitions are taken into account

- $\epsilon - c(q_0) = \{q_0, q_1, q_4\}$
- $\epsilon - c(q_1) = \{q_1, q_4\}$
- $\epsilon - c(q_2) = \{q_2, q_3\}$
- $\epsilon - c(q_3) = \{q_3\}$
- $\epsilon - c(q_4) = \{q_4\}$
- $\epsilon - c(q_5) = \{q_3, q_5\}$

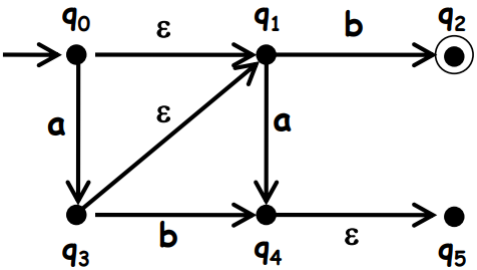


We obtain the set $\{q \mid \epsilon - c(q) \cap T \neq \varnothing\}$, so the accepting states become
$T' = T \cup \{q \mid \epsilon - c(q) \cap T \neq \varnothing\}$

Automata with ϵ-Transitions

To determine the accepting states, ϵ-transitions are taken into account

- $\epsilon - c(q_0) = \{q_0, q_1\}$
- $\epsilon - c(q_1) = \{q_1\}$
- $\epsilon - c(q_2) = \{q_2\}$
- $\epsilon - c(q_3) = \{q_1, q_3\}$
- $\epsilon - c(q_4) = \{q_4, q_5\}$
- $\epsilon - c(q_5) = \{q_4, q_5\}$

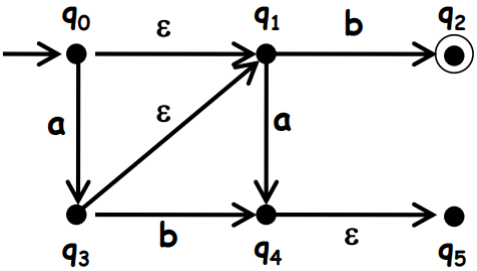


We obtain the set $\{q \mid \epsilon - c(q) \cap T \neq \varnothing\}$, so the accepting states become
$T' = T \cup \{q \mid \epsilon - c(q) \cap T \neq \varnothing\}$

Automata with ϵ–Transitions

To determine the accepting states, ϵ–transitions are taken into account

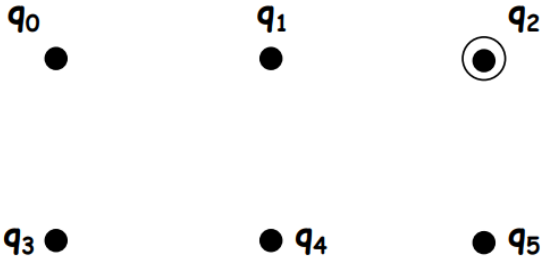
- $\epsilon - c(q_0) = \{q_0, q_1\}$
- $\epsilon - c(q_1) = \{q_1\}$
- $\epsilon - c(q_2) = \{q_2\}$
- $\epsilon - c(q_3) = \{q_1, q_3\}$
- $\epsilon - c(q_4) = \{q_4, q_5\}$
- $\epsilon - c(q_5) = \{q_4, q_5\}$



In this case $T' = \{q_2\}$

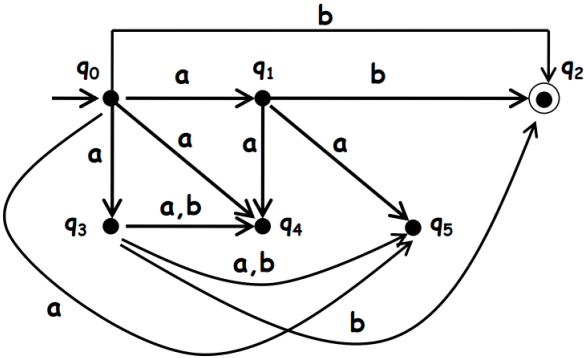
Automata with ϵ-Transitions

Δ	a	b
q_0	$\{q_1, q_3, q_4, q_5\}$	$\{q_2\}$
q_1	$\{q_4, q_5\}$	$\{q_2\}$
q_2	$\varnothing$	$\varnothing$
q_3	$\{q_4, q_5\}$	$\{q_2, q_4, q_5\}$
q_4	$\varnothing$	$\varnothing$
q_5	$\varnothing$	$\varnothing$



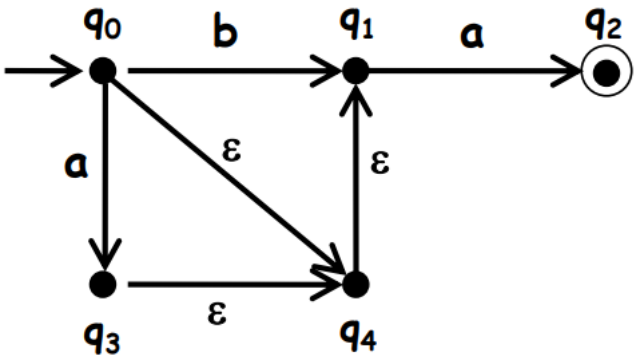
Automata with ϵ-Transitions

Δ	a	b
q_0	$\{q_1, q_3, q_4, q_5\}$	$\{q_2\}$
q_1	$\{q_4, q_5\}$	$\{q_2\}$
q_2	$\varnothing$	$\varnothing$
q_3	$\{q_4, q_5\}$	$\{q_2, q_4, q_5\}$
q_4	$\varnothing$	$\varnothing$
q_5	$\varnothing$	$\varnothing$



Automata with ϵ-Transitions

Design an NFA without ϵ-transitions equivalent to the one shown below:



Automata with ϵ-Transitions

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1,q_4\}$
- $d(\varepsilon\text{-}c(q_0),a)=\{q_2,q_3\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),a))=\{q_1,q_2,q_3,q_4\}$

$\underline{\Delta(q_0,a)=\{q_1,q_2,q_3,q_4\}}$

- $\varepsilon\text{-}c(q_1)=\{q_1\}$
- $d(\varepsilon\text{-}c(q_1),a)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),a))=\{q_2\}$

$\underline{\Delta(q_1,a)=\{q_2\}}$

- $\varepsilon\text{-}c(q_2)=\{q_2\}$
- $d(\varepsilon\text{-}c(q_2),a)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),a))=\varnothing$

$\underline{\Delta(q_2,a)=\varnothing}$

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1,q_4\}$
- $d(\varepsilon\text{-}c(q_0),b)=\{q_1\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),b))=\{q_1\}$

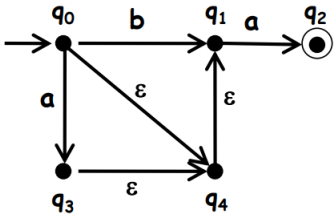
$\underline{\Delta(q_0,b)=\{q_1\}}$

- $\varepsilon\text{-}c(q_1)=\{q_1\}$
- $d(\varepsilon\text{-}c(q_1),b)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),b))=\varnothing$

$\underline{\Delta(q_1,b)=\varnothing}$

- $\varepsilon\text{-}c(q_2)=\{q_2\}$
- $d(\varepsilon\text{-}c(q_2),b)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),b))=\varnothing$

$\underline{\Delta(q_2,b)=\varnothing}$



Automata with ϵ-Transitions

- $\varepsilon\text{-}c(q_3)=\{q_1,q_3,q_4\}$
- $d(\varepsilon\text{-}c(q_3),a)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_3),a))=\{q_2\}$

$\underline{\Delta(q_3,a)=\{q_2\}}$

- $\varepsilon\text{-}c(q_4)=\{q_1,q_4\}$
- $d(\varepsilon\text{-}c(q_4),a)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_4),a))=\{q_2\}$

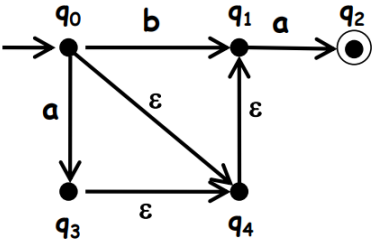
$\underline{\Delta(q_4,a)=\{q_2\}}$

- $\varepsilon\text{-}c(q_3)=\{q_1,q_3,q_4\}$
- $d(\varepsilon\text{-}c(q_3),b)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_3),b))=\varnothing$

$\underline{\Delta(q_3,b)=\varnothing}$

- $\varepsilon\text{-}c(q_4)=\{q_1,q_4\}$
- $d(\varepsilon\text{-}c(q_4),b)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_4),b))=\varnothing$

$\underline{\Delta(q_4,b)=\varnothing}$



Automata with ϵ-Transitions

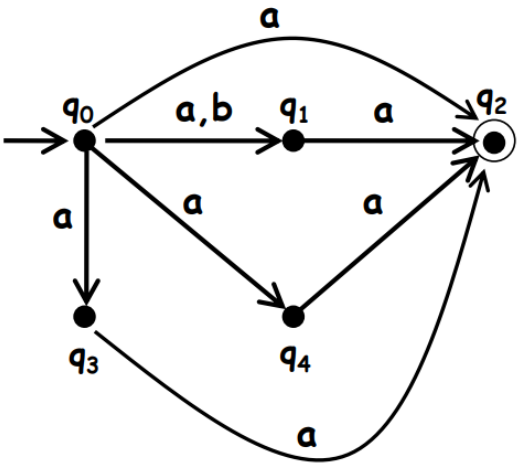
To determine the accepting states T':

- $\epsilon - c(q_0) = \{q_0, q_1, q_4\}$
- $\epsilon - c(q_1) = \{q_1\}$
- $\epsilon - c(q_2) = \{q_2\}$
- $\epsilon - c(q_3) = \{q_1, q_3, q_4\}$
- $\epsilon - c(q_4) = \{q_1, q_4\}$

Therefore $T' = T \cup \{q \mid \epsilon - c(q) \cap T \neq \varnothing\} = \{q_2\} \cup \varnothing = \{q_2\}$

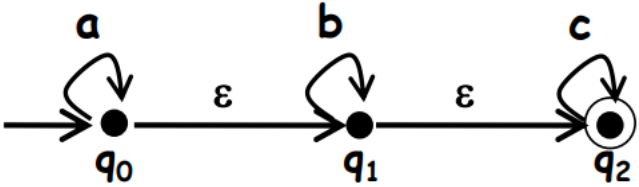
Automata with ϵ-Transitions

Δ	a	b
q_0	$\{q_1, q_2, q_3, q_4\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\varnothing$
q_2	$\varnothing$	$\varnothing$
q_3	$\{q_2\}$	$\varnothing$
q_4	$\{q_2\}$	$\varnothing$



Automata with ϵ-Transitions

Design an NFA without ϵ-transitions equivalent to the one shown below:



Automata with ϵ-Transitions

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_0),a)=\{q_0\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),a))=\{q_0,q_1,q_2\}$

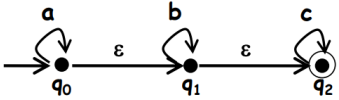
$\underline{\Delta(q_0,a)=\{q_0,q_1,q_2\}}$

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_0),b)=\{q_1\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),b))=\{q_1,q_2\}$

$\underline{\Delta(q_0,b)=\{q_1,q_2\}}$

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_0),c)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),c))=\{q_2\}$

$\underline{\Delta(q_0,c)=\{q_2\}}$



- $\varepsilon\text{-}c(q_1)=\{q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_1),a)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),a))=\varnothing$

$\underline{\Delta(q_1,a)=\varnothing}$

- $\varepsilon\text{-}c(q_1)=\{q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_1),b)=\{q_1\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),b))=\{q_1,q_2\}$

$\underline{\Delta(q_1,b)=\{q_1,q_2\}}$

- $\varepsilon\text{-}c(q_1)=\{q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_1),c)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),c))=\{q_2\}$

$\underline{\Delta(q_1,c)=\{q_2\}}$

- $\varepsilon\text{-}c(q_2)=\{q_2\}$
- $d(\varepsilon\text{-}c(q_2),a)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),a))=\varnothing$

$\underline{\Delta(q_2,a)=\varnothing}$

- $\varepsilon\text{-}c(q_2)=\{q_2\}$
- $d(\varepsilon\text{-}c(q_2),b)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),b))=\varnothing$

$\underline{\Delta(q_2,b)=\varnothing}$

- $\varepsilon\text{-}c(q_2)=\{q_2\}$
- $d(\varepsilon\text{-}c(q_2),c)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),c))=\{q_2\}$

$\underline{\Delta(q_2,c)=\{q_2\}}$

Automata with ϵ–Transitions

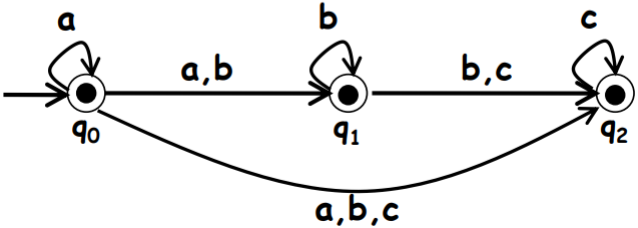
To determine the accepting states T':

- $\epsilon - c(q_0) = \{q_0, q_1, q_2\}$
- $\epsilon - c(q_1) = \{q_1, q_2\}$
- $\epsilon - c(q_2) = \{q_2\}$

Therefore $T' = T \cup \{q \mid \epsilon - c(q) \cap T \neq \varnothing\} = \{q_0, q_1, q_2\}$

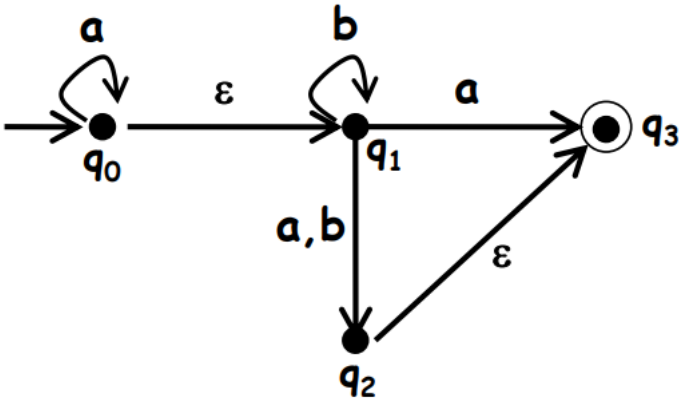
Automata with ϵ-Transitions

Δ	a	b	c
q_0	$\{q_0,q_1,q_2\}$	$\{q_1,q_2\}$	$\{q_2\}$
q_1	$\varnothing$	$\{q_1,q_2\}$	$\{q_2\}$
q_2	$\varnothing$	$\varnothing$	$\{q_2\}$



Automata with ϵ-Transitions

Design an NFA without ϵ-transitions equivalent to the one shown below:



Automata with ϵ-Transitions

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1\}$
- $d(\varepsilon\text{-}c(q_0),a)=\{q_0,q_2,q_3\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),a))=\{q_0, q_1,q_2,q_3\}$

$\underline{\Delta(q_0,a)=\{q_0,q_1,q_2,q_3\}}$

- $\varepsilon\text{-}c(q_1)=\{q_1\}$
- $d(\varepsilon\text{-}c(q_1),a)=\{q_2,q_3\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),a))=\{q_2,q_3\}$

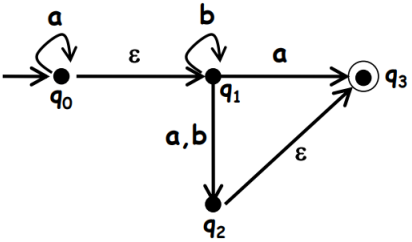
$\underline{\Delta(q_1,a)=\{q_2,q_3\}}$

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1\}$
- $d(\varepsilon\text{-}c(q_0),b)=\{q_1,q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),b))=\{q_1,q_2,q_3\}$

$\underline{\Delta(q_0,b)=\{q_1,q_2,q_3\}}$

- $\varepsilon\text{-}c(q_1)=\{q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_1),b)=\{q_1,q_2,q_3\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),b))=\{q_2\}$

$\underline{\Delta(q_1,b)=\{q_1,q_2,q_3\}}$



Automata with ϵ-Transitions

- $\varepsilon\text{-}c(q_2)=\{q_2,q_3\}$
- $d(\varepsilon\text{-}c(q_2),a)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),a))=\varnothing$

$\underline{\Delta(q_2,a)=\varnothing}$

- $\varepsilon\text{-}c(q_3)=\{q_3\}$
- $d(\varepsilon\text{-}c(q_3),a)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_3),a))=\varnothing$

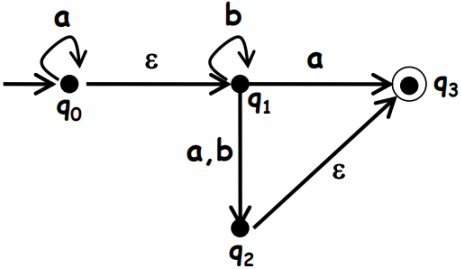
$\underline{\Delta(q_3,a)=\varnothing}$

- $\varepsilon\text{-}c(q_2)=\{q_2,q_3\}$
- $d(\varepsilon\text{-}c(q_2),b)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),b))=\varnothing$

$\underline{\Delta(q_2,b)=\varnothing}$

- $\varepsilon\text{-}c(q_3)=\{q_3\}$
- $d(\varepsilon\text{-}c(q_3),b)=\varnothing$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_3),b))=\varnothing$

$\underline{\Delta(q_3,b)=\varnothing}$



Automata with ϵ–Transitions

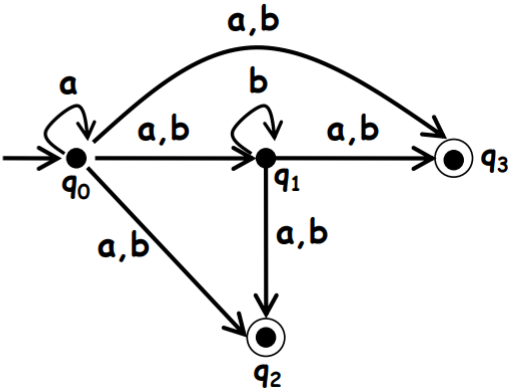
To determine the accepting states T':

- $\epsilon - c(q_0) = \{q_0, q_1\}$
- $\epsilon - c(q_1) = \{q_1\}$
- $\epsilon - c(q_2) = \{q_2, q_3\}$
- $\epsilon - c(q_3) = \{q_3\}$

Therefore $T' = T \cup \{q \mid \epsilon - c(q) \cap T \neq \varnothing\} = \{q_2, q_3\}$

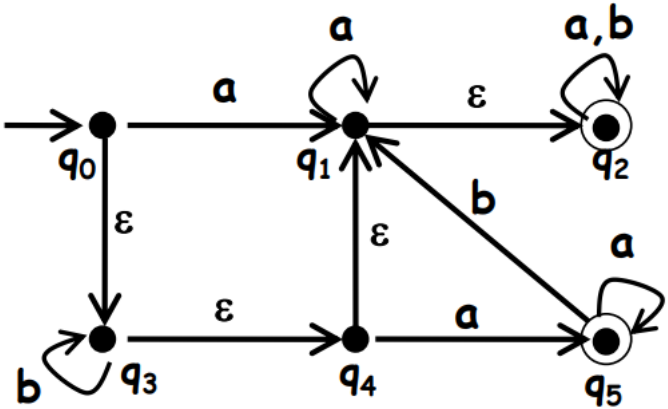
Automata with ϵ-Transitions

Δ	a	b
q_0	$\{q_0, q_1, q_2, q_3\}$	$\{q_1, q_2, q_3\}$
q_1	$\{q_2, q_3\}$	$\{q_2\}$
q_2	$\varnothing$	$\varnothing$
q_3	$\varnothing$	$\varnothing$



Automata with ϵ-Transitions

Design an NFA without ϵ-transitions equivalent to the one shown below:



Automata with ϵ-Transitions

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1,q_2,q_3,q_4\}$
- $d(\varepsilon\text{-}c(q_0),a)=\{q_1,q_2,q_5\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),a))=\{q_1,q_2,q_5\}$

$\underline{\Delta(q_0,a)=\{q_1,q_2,q_5\}}$

- $\varepsilon\text{-}c(q_1)=\{q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_1),a)=\{q_1,q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),a))=\{q_1,q_2\}$

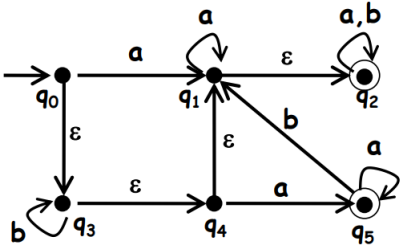
$\underline{\Delta(q_1,a)=\{q_1,q_2\}}$

- $\varepsilon\text{-}c(q_0)=\{q_0,q_1,q_2,q_3,q_4\}$
- $d(\varepsilon\text{-}c(q_0),b)=\{q_2,q_3\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_0),b))=\{q_1,q_2,q_3,q_4\}$

$\underline{\Delta(q_0,b)=\{q_1,q_2,q_3,q_4\}}$

- $\varepsilon\text{-}c(q_1)=\{q_1,q_2\}$
- $d(\varepsilon\text{-}c(q_1),b)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_1),b))=\{q_2\}$

$\underline{\Delta(q_1,b)=\{q_2\}}$



Automata with ϵ-Transitions

- $\varepsilon\text{-}c(q_2)=\{q_2\}$
- $d(\varepsilon\text{-}c(q_2),a)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),a))=\{q_2\}$

$\underline{\Delta(q_2,a)=\{q_2\}}$

- $\varepsilon\text{-}c(q_3)=\{q_1,q_2,q_3,q_4\}$
- $d(\varepsilon\text{-}c(q_3),a)=\{q_1,q_2,q_5\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_3),a))=\{q_1,q_2,q_5\}$

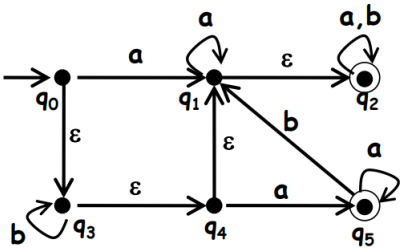
$\underline{\Delta(q_3,a)=\{q_1,q_2,q_5\}}$

- $\varepsilon\text{-}c(q_2)=\{q_2\}$
- $d(\varepsilon\text{-}c(q_2),b)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_2),b))=\{q_2\}$

$\underline{\Delta(q_2,b)=\{q_2\}}$

- $\varepsilon\text{-}c(q_3)=\{q_1,q_2,q_3,q_4\}$
- $d(\varepsilon\text{-}c(q_3),b)=\{q_2,q_3\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_3),b))=\{q_1,q_2,q_3,q_4\}$

$\underline{\Delta(q_3,b)=\{q_1,q_2,q_3,q_4\}}$



Automata with ϵ-Transitions

- $\varepsilon\text{-}c(q_4)=\{q_1,q_2,q_4\}$
- $d(\varepsilon\text{-}c(q_4),a)=\{q_1,q_2,q_5\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_4),a))=\{q_1,q_2,q_5\}$

$\underline{\Delta(q_4,a)=\{q_1,q_2,q_5\}}$

- $\varepsilon\text{-}c(q_5)=\{q_5\}$
- $d(\varepsilon\text{-}c(q_5),a)=\{q_5\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_5),a))=\{q_5\}$

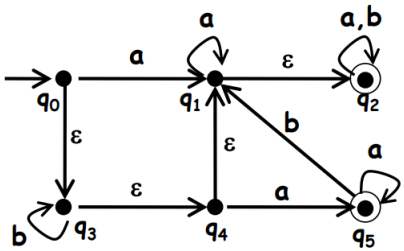
$\underline{\Delta(q_5,a)=\{q_5\}}$

- $\varepsilon\text{-}c(q_4)=\{q_1,q_2,q_4\}$
- $d(\varepsilon\text{-}c(q_4),b)=\{q_2\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_4),b))=\{q_2\}$

$\underline{\Delta(q_4,b)=\{q_2\}}$

- $\varepsilon\text{-}c(q_5)=\{q_5\}$
- $d(\varepsilon\text{-}c(q_5),b)=\{q_1\}$
- $\varepsilon\text{-}c(d(\varepsilon\text{-}c(q_5),b))=\{q_1,q_2\}$

$\underline{\Delta(q_5,b)=\{q_1,q_2\}}$



Automata with ϵ–Transitions

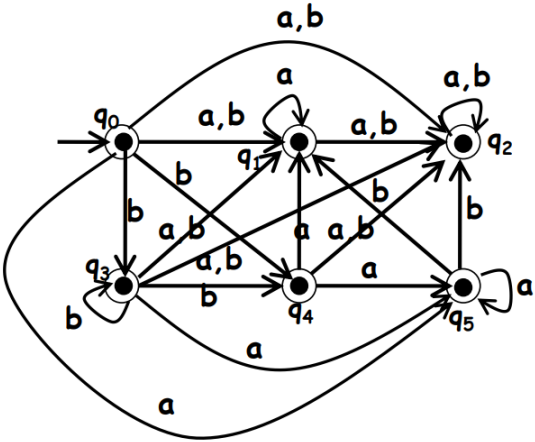
To determine the accepting states T':

- $\epsilon - c(q_0) = \{q_0, q_1, q_2, q_3, q_4\}$
- $\epsilon - c(q_1) = \{q_1, q_2\}$
- $\epsilon - c(q_2) = \{q_2\}$
- $\epsilon - c(q_3) = \{q_1, q_2, q_3, q_4\}$
- $\epsilon - c(q_4) = \{q_1, q_2, q_4\}$
- $\epsilon - c(q_5) = \{q_5\}$

Therefore $T' = T \cup \{q \mid \epsilon - c(q) \cap T \neq \varnothing\} = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

Automata with ϵ-Transitions

Δ	a	b
q_0	$\{q_1,q_2,q_5\}$	$\{q_1,q_2,q_3,q_4\}$
q_1	$\{q_1,q_2\}$	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_2\}$
q_3	$\{q_1,q_2,q_5\}$	$\{q_1,q_2,q_3,q_4\}$
q_4	$\{q_1,q_2,q_5\}$	$\{q_2\}$
q_5	$\{q_5\}$	$\{q_1,q_2\}$



Contenido

1. Automata with ϵ-Transitions

2. Equivalence between NFA-ϵ and NFA (and subsequently DFA)

3. Regular Languages and Finite Automata (Construction)

Automata with ϵ-Transitions

Theorem

For every regular expression R, an NFA-ϵ M can be constructed such that $L(R) = L(M)$

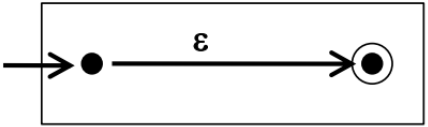
Automata with ϵ-Transitions

Theorem

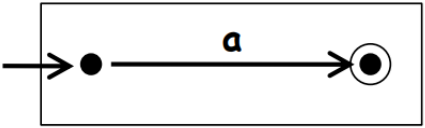
It is proven that given the definition of a regular language, a finite automaton can be constructed

- $\varnothing$ is a regular language
- $\{\epsilon\}$ is a regular language
- For every $a \in \Sigma$, $\{a\}$ is a regular language
- If A and B are regular languages, then $A \cup B$, $A \cdot B$, and A^* are regular languages
- No other language is regular

Automata with ϵ-Transitions

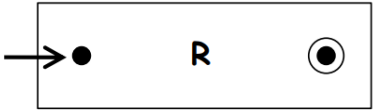


Automaton whose accepted language is the empty string, $\{\epsilon\}$

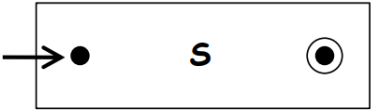


Automaton whose accepted language is the string a, $\{a\}$

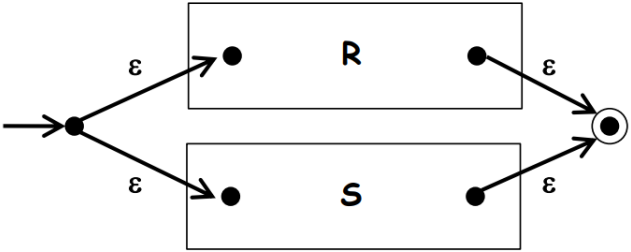
Automata with ϵ-Transitions



Autómata que acepta R

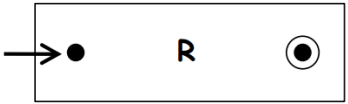


Autómata que acepta S

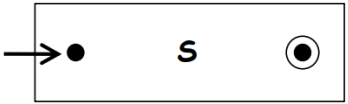


Autómata que acepta R∪S

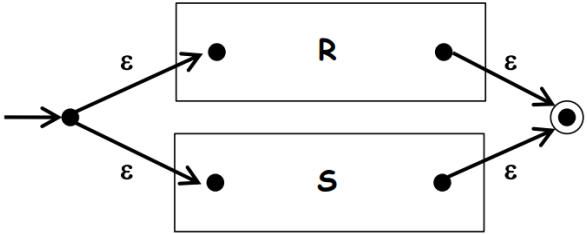
Automata with ϵ-Transitions



Autómata que acepta R



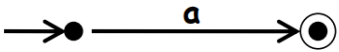
Autómata que acepta S



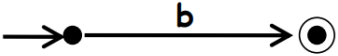
Autómata que acepta R∪S

Construya un autómata que acepte a∪b

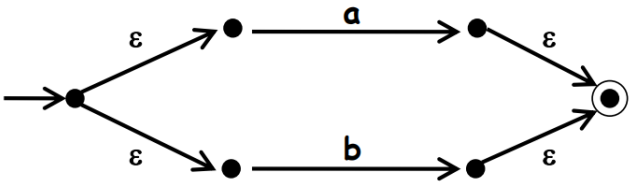
Automata with ϵ-Transitions



Autómata que acepta a

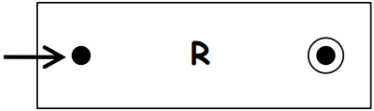


Autómata que acepta b

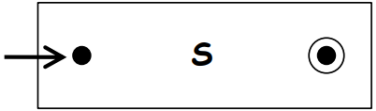


Autómata que acepta a∪b

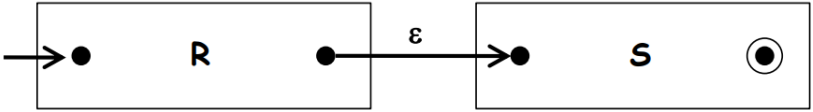
Automata with ϵ-Transitions



Autómata que acepta R

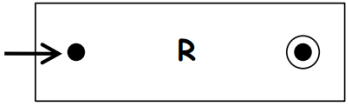


Autómata que acepta S

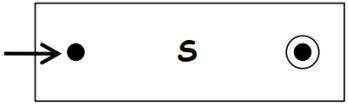


Autómata que acepta R·S

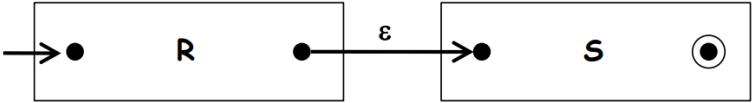
Automata with ϵ-Transitions



Autómata que acepta R



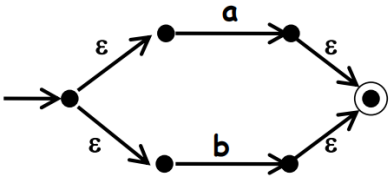
Autómata que acepta S



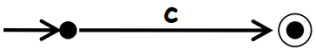
Autómata que acepta R·S

Construya un autómata que acepte (a∪b)·c

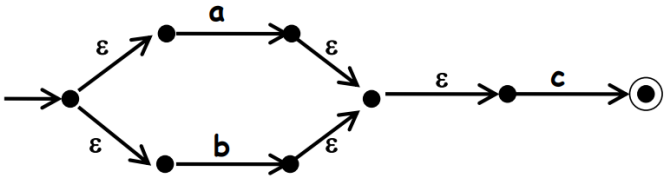
Automata with ϵ-Transitions



Autómata que acepta a∪b



Autómata que acepta c



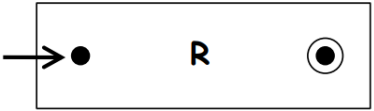
Autómata que acepta (a∪b)·c

Automata with ϵ-Transitions

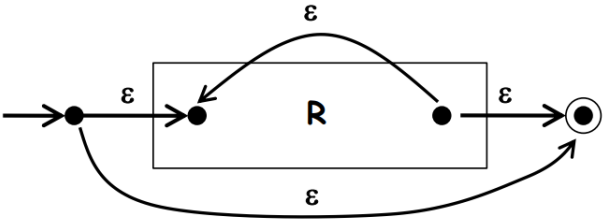
Construct an automaton for each of the following regular expressions:

- $ab \cup a$
- $(a \cup b)(a \cup b)$
- $a(a \cup b)b$

Automata with ϵ-Transitions

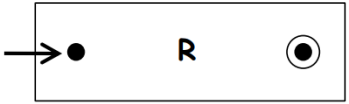


Autómata que acepta R

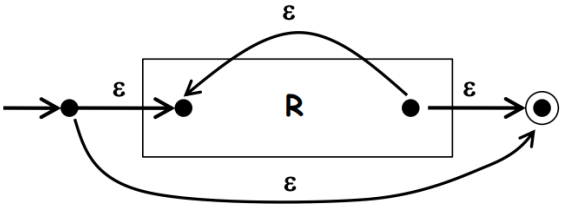


Autómata que acepta R*

Automata with ϵ-Transitions



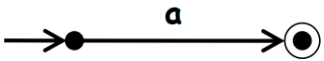
Autómata que acepta R



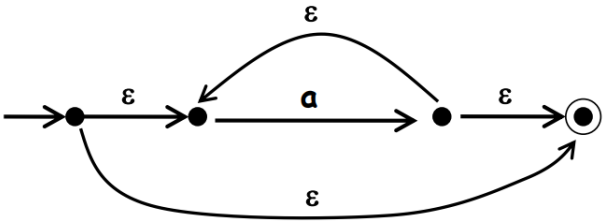
Autómata que acepta R*

Construya un autómata

Automata with ϵ-Transitions



Autómata que acepta a



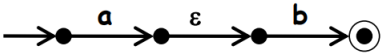
Autómata que acepta R*

Automata with ϵ-Transitions

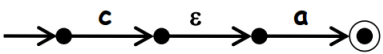
Construct the automaton that recognizes $ab \cup ca$

Automata with ϵ-Transitions

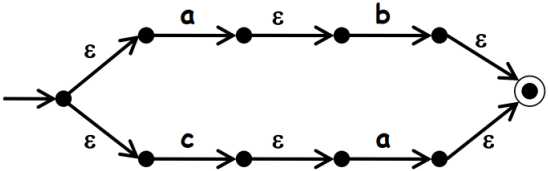
Construct the automaton that recognizes $ab \cup ca$



Autómata que acepta ab



Autómata que acepta ca



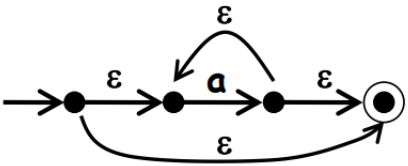
Autómata que acepta ab∪ca

Automata with ϵ-Transitions

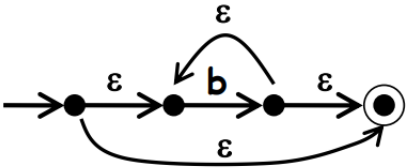
Construct the automaton that recognizes $a^*b \cup b^*a$

Automata with ϵ-Transitions

Construct the automaton that recognizes $a^*b \cup b^*a$



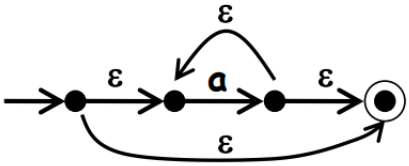
Autómata que acepta a*



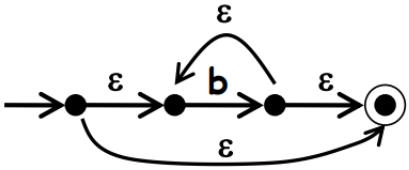
Autómata que acepta b*

Automata with ϵ-Transitions

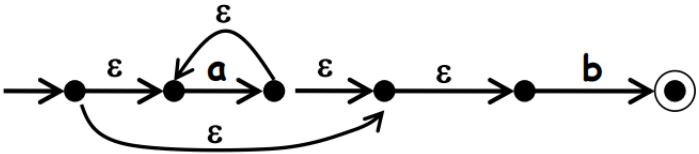
Construct the automaton that recognizes $a^*b \cup b^*a$



Autómata que acepta a*



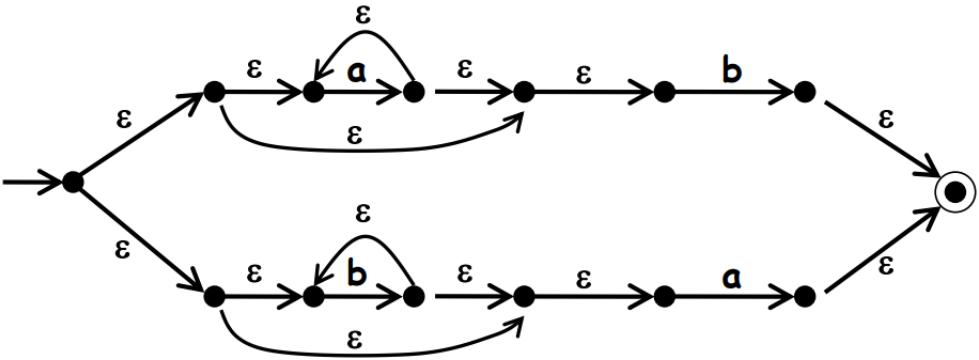
Autómata que acepta b*



Autómata que acepta a*b

Automata with ϵ-Transitions

Construct the automaton that recognizes $a^*b \cup b^*a$



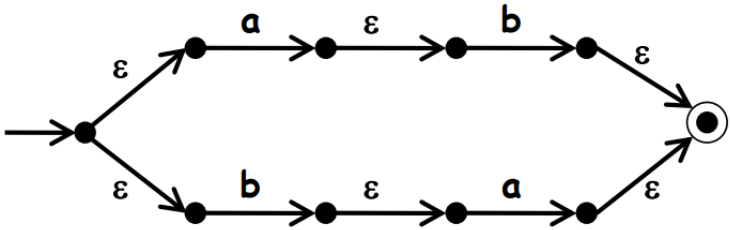
Autómata que acepta a*b∪b*a

Automata with ϵ-Transitions

Construct the automaton that recognizes $(ab \cup ba)^*$

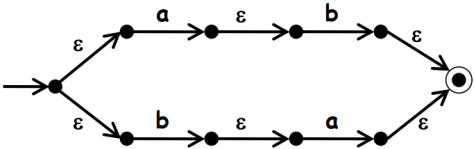
Automata with ϵ-Transitions

Construct the automaton that recognizes $(ab \cup ba)^*$

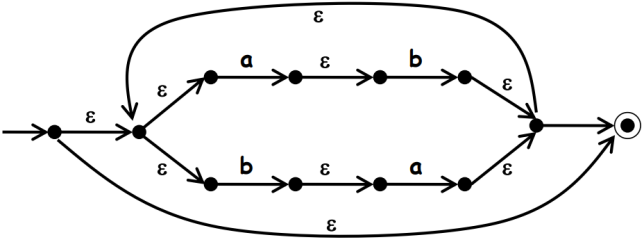


Autómata que acepta (ab∪ba)

Automata with ϵ-Transitions



Autómata que acepta (ab∪ba)



Autómata que acepta (ab∪ba)*

Automata with ϵ-Transitions

Construct the automaton that recognizes $(ab)(ab)^* \cup b^*$, showing the construction of the automata that accept the languages given by the following regular expressions:

- a
- b
- ab
- $(ab)^*$
- b^*
- $(ab)(ab)^*$
- $(ab)(ab)^* \cup b^*$

References

Kozen, D. C. (2007)
Automata and computability
Springer Science & Business Media. Lectures 3–5.